

Persistence Is Not Accumulation: Erasure, Suppression, and Re-instatement in Online In-Weight Memory for Language Models

Xuhao Lin

Independent researcher linxuhao84@gmail.com

July 2026 — v2 (preprint; v1 archived as doi:10.5281/zenodo.21199026)

Abstract

Can a frozen language model accumulate new facts *in its weights*, one at a time, as they arrive? We study single-pass online fact streams written into a LoRA adapter on a frozen base, with a de-confounded instrument: per-fact recall timelines, a two-alternative recognition probe that separates *storage* from *expression*, matched compute budgets, and a capability firewall (adapter-off equals base) verified in every run, on two substrates (Qwen3.5-2B and SmolLM2-1.7B-Instruct). Four findings. (1) **Accumulation is re-instatement, not persistence**: median time-to-first-miss is nearly identical (3–8 writes) for every mechanism we test, while end-of-stream retention varies by 12×; what separates mechanisms is the probability that a missed fact is later recovered (4%–94%). Half-life is the wrong abstraction for online memory. (2) **Whether forgetting is erasure or suppression depends on the write protection**: facts forgotten under naked sequential writes are barely recognizable (2-AFC at or near chance), whereas facts forgotten under accumulated elastic-weight consolidation (EWC) remain largely recognizable — on the second substrate EWC recalls *nothing* while recognizing well above chance, suppression in its purest form. (3) **Error-gated replay converts suppressed facts into expressed ones at zero extra budget**: spending an unchanged 4-step replay budget on self-test failures first helps every replay-bearing mechanism on both substrates, and lifts the EWC+replay composition to **85–96%** of a 48-fact day recallable from weights alone ($n=5$ seeds) — provided the self-tests are fresh: the gain shows a clean dose–response in self-test cadence and vanishes by one test per 12 writes. (4) **The depth gradient of online writes is an expression gradient, not a storage gradient**: recall capacity concentrates in the last third of the stack (opposite to the early-layer placement reported for offline batch injection), yet recognition shows the middle third *stores* nearly as much as the full stack while expressing a quarter of it. Results are multi-seed directions with per-seed numbers throughout; we release the instrument and raw per-fact timelines.

1 Introduction

Language-model agents increasingly ship with external memory — files, retrieval, context — while their weights stay frozen after deployment. A recurring alternative is to let the agent *write* to a parameter-efficient adapter online, one interaction at a time, and consolidate at rest. Recent systematic work has audited LoRA-as-memory in the *batch* regime: inject a corpus by multi-epoch fine-tuning, measure capacity, saturation, and composition (Back et al., 2026). That

audit explicitly defers the regime that deployed agents actually face: a *time-incremental stream*, one fact per turn, single pass, no i.i.d. shuffling.

This paper is an empirical audit of that streaming regime. The regime is harsh by construction: sequential SGD has no mechanism for preserving earlier writes, and interference accumulates. But *how* it is harsh turns out to be structured, and the structure is invisible to the aggregate accuracy numbers that continual-learning papers usually report. Our instrument therefore records *per-fact* trajectories: when each fact is first recalled, when it first fails, whether it ever comes back, and — separately — whether it remains *recognizable* when it is no longer *recallable*.

Contributions.

1. **A de-confounded streaming instrument** (§3): 48 collision-free facts written one per turn into a LoRA on a frozen base; greedy-recall and 2-AFC recognition probes; a pinned recovery statistic; compute-matched mechanism comparisons; and a capability firewall (adapter-off GSM8K vs. base) checked in every run (+0.00 in all runs, both substrates, all mechanisms and seeds).
2. **Accumulation is re-instatement** (§4.1): across mechanisms spanning $12\times$ in final retention, median time-to-first-miss varies only from 3 to 8 subsequent writes. Mechanisms differ almost entirely in recovery-after-miss (4%–94%). Survival-time (“half-life”) framings of online memory measure the wrong thing.
3. **Erasure vs. suppression is mechanism-dependent** (§4.2): recognition probes show naked-SGD forgetting destroys fact-specific binding (2-AFC at or near chance), while EWC forgetting leaves most facts discriminable behind a failing recall readout — on SmolLM2-1.7B-Instruct, EWC recalls 0/48 while recognizing well above chance. This connects the machine-unlearning observation that “forgotten” knowledge is often latent (Xu et al., 2025; Zhang et al., 2025; Kotha et al., 2024) to online continual learning, with a mechanism-level answer to *when* each case obtains.
4. **Error-gated replay exploits suppression at zero extra cost — if the self-tests are fresh** (§4.3): with the replay budget held fixed, selecting replay targets by self-test failure helps every replay-bearing arm on both substrates and lifts the EWC+replay composition to 41–46/48 (85–96%, $n=5$). The gain decays monotonically with self-test cadence ($44.2 \rightarrow 35.3 \rightarrow 32.0$ at one test per 2/6/12 writes) — staleness converts targeting back into uniformity. We do not claim the selection rule as novel (Aljundi et al., 2019; Hazard et al., 2025; Lu et al., 2026); the contributions are the budget-matched decomposition and the cadence dose–response.
5. **The depth gradient is an expression gradient** (§4.4): at matched adapter capacity, *recall* concentrates in the last third of the stack (the late band alone matches full-stack replay recall) while the first third recalls almost nothing — opposite to the early-layer/FFN advantage reported for batch injection (Back et al., 2026). Recognition, however, shows mid-band *storage* near full-stack levels: the single-pass regime chiefly limits *readout depth*, which partially reconciles the two regimes.

We present these as *floors and directions*, not ceilings and points: two small substrates, synthetic facts, 3–5 seeds per condition on nondeterministic hardware (§6).

2 Related work

LoRA as knowledge memory (batch regime). Back et al. (2026) systematically audit LoRA as a parametric memory under multi-epoch injection: capacity scales with rank and saturates; QA-format supervision dominates raw text; multi-LoRA composition is bottlenecked by routing and merging. Their stated next step — time-incremental streams where information “must be stored, retrieved, and revised continuously” — is precisely the regime studied here, and at least one of their operational conclusions (early-layer placement) inverts for *recall* in it (§4.4). **Continual learning and replay selection.** Catastrophic forgetting under sequential training is classical (McCloskey & Cohen, 1989; Kirkpatrick et al., 2017). Replay selection by estimated interference (MIR; Aljundi et al. 2019), encoding-time surprise (SuRe; Hazard et al. 2025), forgetting-curve schedules (MSSR; Lu et al. 2026), and prioritized experience replay (Schaul et al., 2015) all improve on uniform replay; our miss-gated rule is a member of this family, applied via generative self-testing, and we claim only its budget-matched interaction with the substrate and its cadence dependence. Orthogonal-projection methods (Farajtabar et al., 2020) and pseudo-rehearsal (Robins, 1995) bound the design space we sweep. **Forgetting as suppression.** In machine unlearning, “forgotten” knowledge is frequently recoverable by quantization, benign relearning, or prompting (Zhang et al., 2025; Xu et al., 2025; Kotha et al., 2024); example-level forgetting-and-relearning events are pervasive even in offline training (Toneva et al., 2019). We import the storage/expression distinction (Allen-Zhu & Li, 2024) into the online regime with a per-fact recognition meter. **Memory architectures.** Complementary-systems designs — fast weights, sleep-time consolidation, activation-state hippocampi (Behrouz et al., 2024; Fang et al., 2025) — motivate but are orthogonal to this audit: we characterize the write dynamics such architectures inherit.

3 The instrument

Substrates. Primary: Qwen3.5-2B (24 layers). Transfer: SmolLM2-1.7B-Instruct (Allal et al., 2025) — a different organization, architecture family, and training corpus — run with *all hyperparameters unchanged* from the primary substrate, so transfer tests the qualitative structure, not tuned performance. Both fp32, frozen; one LoRA adapter (rank 64, $\alpha=128$, all linear modules unless stated) is the sole trainable memory. AdamW, lr 3×10^{-5} , gradient clip 1.0. Hardware: consumer ROCm GPUs; runs are nondeterministic even at fixed seed, so every number below is reported per-seed (seeds 1234/2025/777, plus 7/42 where $n=5$) and interpreted as a direction.

Stream. $N=48$ facts of the form “*The user’s {attribute} is {value}.*” with unique attributes and unique three-syllable pseudoword values (collision-free; unguessable; answers cannot be produced without the write). One fact per turn, single pass: 8 gradient steps on the fact (loss on value tokens only), then the mechanism’s bookkeeping, then (for replay mechanisms) exactly $m=4$ replay gradient steps on stored past facts. Budgets are identical across mechanisms.

Mechanisms. *naked*: writes only. *BF*: Benna–Fusi multi-timescale synaptic cascade (Benna & Fusi, 2016) applied to adapter parameters. *EWC*: elastic weight consolidation (Kirkpatrick et al., 2017) in the accumulating regime — diagonal Fisher of each fact is *summed* into a running Fisher (never rebuilt), anchor re-set after each fact, $\lambda=300$ (results are λ -robust across 100–1000

mechanism	final recall /48 (seeds)	mean	median first-miss	recovery	recognized /48
naked	4 / 4 / 3	3.7	3	4%	33 / 23 / 26
BF cascade [†]	4 / 3 / 5	4.0	5	10%	—
EWC ($\lambda 300$)	5 / 18 / 9	10.7	5	27%	38 / 45 / 38
replay (uniform)	24 / 16 / 17	19.0	5	76%	45 / 40 / 45
replay (miss)	15 / 28 / 27	23.3	4	87%	45 / 46 / 46
ewc+replay (uniform), $n=5$	32 / 42 / 42 / 39 / 29	36.8	8	81%	46–48
ewc+replay (miss), $n=5$	46 / 44 / 46 / 44 / 41	44.2	7	94%	48 × 5

Table 1: Qwen3.5-2B, 48-fact single-pass stream, probes every 2 writes. Recovery is pooled over seeds (pinned definition, §3); recognition is the final-probe 2-AFC count (chance = 24); a strict margin threshold (> 1 nat) changes no count by more than 4. [†]BF numbers are from pre-recognition-meter runs (3-seed finals shown; the released fine-probe pair gives 3/6, recovery 10%); its rank/timescale sweep is flat (2–5/48) across $dt \in \{0.1, 0.5\}$, cascade depth $\{8, 32\}$. Firewall: +0.00 in every run. All rows reproducible from the released timelines via `analyze.py`.

on the primary substrate). *replay*: 4 steps/turn on buffer samples, uniform. *ewc+replay*: both. *Selection policy*: uniform, or *miss* — replay targets are drawn first from facts that failed the most recent self-test, filled uniformly. The self-test is the recall probe itself; the information it uses (which stored facts currently fail) is available to any deployed agent that keeps a log of what it wrote. Self-test cadence is a mechanism hyperparameter; default one test per 2 writes, swept in §4.3.

Probes. Every 2 writes, all facts so far are probed. *Recall*: greedy completion of the cloze prefix; substring match on the value. *Recognition*: sum log-probability margin of the true value vs. a fixed distractor value sampled from the *same stream* — the distractor was written into the same adapter with the same procedure, so a positive margin measures fact-specific binding, not token familiarity. Chance is 24/48. *Recovery* (pinned definition): a fact is a recovery candidate if it was recalled at some probe and missed at a later one; it recovered if it is recalled again after its first miss. *Firewall*: GSM8K (10-item fixed subset) with the adapter disabled, before and after the stream; any deviation from the base score indicates capability damage.

4 Results

4.1 Accumulation is re-instatement, not persistence

Table 1 spans a $12\times$ range in final retention (3.7 to 44.2 of 48). If mechanisms differed by making facts *persist* longer, time-to-first-miss would covary with that range. It does not: the median fact first fails 3–8 writes after acquisition *under every mechanism*. What covaries with retention is what happens *after* the first miss: under naked writes 4% of missed facts ever return; under uniform replay 76%; under the error-gated composition 94%. End-of-stream alive counts tell the same story: naked 11/144, EWC 32/144, replay 57/144, composition 184/240, error-gated composition 221/240.

Online in-weight memory in this regime is therefore not a leaky tank whose drain rate mechanisms adjust; it is a *flickering* store in which nearly everything falls below the recall threshold quickly, and capacity is determined by the re-instatement process. Aggregate accuracy and half-life statistics — the standard reporting in this literature — cannot see this structure;

per-fact timelines can. The same dynamic is known *inside* offline training as example forgetting events (Toneva et al., 2019): shuffled multi-epoch training is, in effect, a built-in re-instatement schedule. Our results show the online regime obeys the same law, with the schedule made explicit and paid for.

4.2 Forgetting is erasure or suppression, depending on the write rule

The recognition column of Table 1 dissociates two kinds of forgetting. Under naked writes, recognition of the full stream is at or near chance (23–33 vs. chance 24 on Qwen3.5-2B; 29–30 on SmolLM2-1.7B-Instruct): when recall of a fact dies, its fact-specific binding is largely gone — *erasure*. Under accumulated EWC, recall is poor but recognition is far above chance: 38–45 of 48 ($\sim 84\%$) on Qwen3.5-2B at 10.7 recalled, and — the purest case — **0/48 recalled with 31–38 recognized on SmolLM2-1.7B-Instruct**: everything the readout lost, the weights still hold — *suppression* below the generative readout, not destruction. Replay-based mechanisms recognize 40–48 while recalling 16–46; the error-gated composition recognizes *everything* (48×5) while recalling 41–46.

This gives a mechanism-level answer to a question the unlearning literature has raised from the other side — “is forgotten knowledge deleted or hidden?” (Xu et al., 2025; Zhang et al., 2025): *it depends on what protected the weights while subsequent writes arrived*. Unprotected interference deletes; Fisher-anchored interference hides. The practical consequence is §4.3.

4.3 Error-gated replay: same budget, large gains — if the self-tests are fresh

Holding the replay budget fixed at 4 steps/turn and changing only *selection* — replay facts that failed the agent’s most recent self-test first — helps every replay-bearing arm on both substrates: Qwen3.5-2B replay 19.0 $\rightarrow$ 23.3, composition 36.8 $\rightarrow$ **44.2** ($n=5$ each, distributions barely overlapping: miss minimum 41 vs. uniform maximum 42); SmolLM2-1.7B-Instruct replay 26.0 $\rightarrow$ 32.3, composition 22.7 $\rightarrow$ 36.0. Recovery-after-miss rises to 87–94%. In v1 of this paper, with fewer seeds and one substrate, we read the plain-replay gain as noise and called the effect conditional on the EWC substrate; the fuller data revise that: **miss-gating helps wherever a latent pool exists (i.e., every protected mechanism), and helps the compositions most** (+7.4/+13.3 vs. +4.3/+6.3). The mechanism reading — suppressed misses are near-threshold and therefore cheap to re-instate, and recognition shows every replay-bearing arm carries a large latent pool — is consistent with all eight comparisons but is an interpretation, not a demonstrated law.

Cadence dose-response. Because the miss list comes from self-tests, its freshness is a resource. Sweeping self-test cadence on the winning composition (Qwen3.5-2B, 3 seeds): one test per 2 writes $\rightarrow$ **44.2** (recovery 94%); per 6 writes $\rightarrow$ 35.3 (75%) — statistically indistinguishable from uniform replay at 36.8; per 12 writes $\rightarrow$ 32.0 (53%). Stale miss lists degrade targeting back to uniformity. The candidate design rule our results support is therefore cadence-qualified: **pair a suppression-inducing protector with error-gated re-instatement driven by frequent self-tests** — in this setting, roughly one self-test per two writes. In cognitive terms the composition implements Leitner-style spaced repetition with retrieval practice (Roediger & Karpicke, 2006) in weight space; the freshness requirement is the part the flash-card analogy does predict (expanding-interval schedules assume you *know* what you just failed).

	early (0–7)	mid (8–15)	late (16–23)	all 24
naked recall	1.0	2.3	2.7	3.7
EWC recall	1.0 (0%)	3.0 (2%)	8.0 (29%)	10.7 (27%)
replay recall	3.0 (34%)	11.0 (61%)	21.3 (74%)	19.0 (76%)
ewc+replay recall	—	27.7	26.0	36.8
replay <i>recognized</i>	34.3	42.7	— [‡]	43.3
EWC <i>recognized</i>	28.3	31.3	— [‡]	40.3

Table 2: Mean final recall /48 (3 seeds; recovery-% in parentheses) with the adapter restricted to matched-capacity 8-layer bands (14.5M parameters) vs. the full stack ($\sim$ 43M), and mean final recognition where measured. [‡]Late-band single-mechanism runs predate the recognition meter; the late-band *composition* recognizes 48/48/47 while recalling 26.

mechanism (SmolLM2-1.7B-Instruct)	final recall /48 (seeds)	mean	recovery	recognized /48
naked	5 / 2 / 4	3.7	6%	29 / 29 / 30
EWC (λ 300)	0 / 0 / 0	0.0	0%	31 / 36 / 38
replay (uniform)	26 / 24 / 28	26.0	77%	47 / 42 / 46
replay (miss)	38 / 28 / 31	32.3	88%	47 / 45 / 48
ewc+replay (uniform)	21 / 23 / 24	22.7	78%	48 / 46 / 46
ewc+replay (miss)	33 / 31 / 44	36.0	85%	48 $\times$ 3

Table 3: SmolLM2-1.7B-Instruct with all hyperparameters carried over unchanged from Qwen3.5-2B. Firewall 0.40 $\rightarrow$ 0.40 (adapter-off equals base) on all composition runs.

4.4 The depth gradient is an expression gradient

Restricting the adapter to matched-capacity 8-layer bands (Table 2) yields a monotone depth gradient in *recall*: the early band recalls almost nothing under every mechanism; the late band alone matches full-stack replay recall (21.3 vs. 19.0, with the tightest seed triple in the study, 21/22/21) at a third of the parameters, and it is the only band in which EWC’s suppression-and-recovery behavior exists at all (29% vs. 0–2%). Restricting to the middle band — the region favored by locate-then-edit knowledge editing (Meng et al., 2022, 2023) and by RL layer-contribution analyses (Zhang et al., 2026) — *hurts* recall under every protected mechanism.

Recognition, however, changes the interpretation. The mid band’s replay arm *recognizes* 42.7 of 48 — essentially full-stack level (43.3) — while *recalling* 11.0 of 48, and the late-band *composition* recognizes 48/48/47 while recalling 26: restricted bands can *store* nearly everything and *express* a fraction of it. The early band is a genuine (if partial) storage desert: its protected-arm recognition (28–36) sits closest to chance. So the regime contrast with batch injection (Back et al., 2026) is best stated as a decomposition rather than a contradiction: **single-pass online writes can deposit latent bindings at many depths, but greedy recall of them is readout-proximal**; multi-epoch batch training, which can afford to build early-layer features and re-extract them over many passes, faces no such expression bottleneck. Whether batch-trained early-layer memories show the same storage/expression split is an open question our instrument could answer in their regime.

4.5 Substrate transfer: structure replicates, magnitudes do not

Running the headline arms on SmolLM2-1.7B-Instruct with *no retuning* (Table 3) tests which findings are structural. Replicated: the naked floor; the re-instatement signature (first-miss medians 4–18 writes while recovery spans 0–88%); large latent pools behind every protected mechanism (recognition 42–48 at recall 21–44); miss-gating’s budget-free gain; the miss-composition as the best arm; and the firewall. Not transferred: *magnitudes*. Most sharply, EWC-alone — 11–18% of the stream on Qwen3.5-2B — recalls **zero** on SmolLM2-1.7B-Instruct at the same λ , while recognizing well above chance: the suppression phenomenon at its purest, and a warning that regularization strengths tuned on one substrate can silently zero the readout on another. The uniform composition also *loses* to plain replay here (22.7 vs. 26.0), though the miss-composition recovers the lead (36.0). Claims in this paper should be read at the structure level; per-substrate magnitudes require per-substrate tuning we deliberately did not do.

4.6 The firewall holds everywhere

In every run of every mechanism, band, policy, cadence, and substrate, disabling the adapter returns exactly the base model’s GSM8K score (0.70 $\rightarrow$ 0.70 on Qwen3.5-2B; 0.40 $\rightarrow$ 0.40 on SmolLM2-1.7B-Instruct; fixed 10-item subset). Online writing at any tested intensity costs the frozen base nothing, by construction — the architectural premise (frozen capability, disposable memory cartridges) survives its hardest workload to date.

5 Discussion

For memory-agent design. The numbers argue for a specific division of labor: an external log (“the file”) as ground truth and *syllabus*, and an in-weight day store trained from it by error-gated re-instatement — with the log also supplying frequent self-tests, whose cadence is a first-class budget item. The day store is then best understood not as a transcript but as a *familiarity index*: in the strongest configurations it recognizes 100% of the day while recalling 85–96%, and the recognition surplus is exactly what re-grounding against the log can exploit. **For theory.** A first-order scale-invariance argument says per-write “gentleness” cannot create accumulation (shrinking the step shrinks a fact’s own margin and its interference on others proportionally); only write *geometry* and *re-instatement schedules* can. Our mechanism ladder is a walk along exactly those two levers, and linear-attention fast-weight memories — which meta-learn their write geometry offline (Fang et al., 2025) — can be read as the activation-state solution to the same problem. **For the unlearning debate.** Erasure and suppression are both real and separable by a cheap 2-AFC meter; arguments about whether forgetting “really” deletes should condition on the write-time protection regime. **Beyond one day.** Preliminary multi-day experiments (day-stream plus nightly consolidation into a persistent core, to be reported separately) show the same ordering across the day boundary: full ground-truth re-instatement $\gg$ lossy self-re-instatement $\gg$ none.

6 Limitations

Two small substrates, one hyperparameter setting (tuned on the first); per-substrate magnitudes do not transfer (§4.5) and are floors. Results are directions on nondeterministic ROCm hardware with 3–5 seeds per condition. The stream is synthetic and collision-free; real facts correlate,

conflict, and get revised — revision streams (e.g., CounterFact-style) are future work. Recall probes are form-matched to the writes (this inflates all absolute recall numbers; it is held constant across comparisons); paraphrase probing is future work. The self-test that gates replay is the measurement probe; we characterized its cadence (§4.3) but a measurement-independent self-test would be cleaner. Late-band single-mechanism runs lack recognition data. The firewall subset is small (10 GSM8K items) and near-trivial on SmolLM2-1.7B-Instruct (base 0.40). BF rows predate the recognition meter. Day-scale only: cross-day consolidation is reported separately.

7 Conclusion

On a single-pass fact stream, in-weight memory does not persist — it flickers, and it accumulates exactly to the degree that something re-instates it. Whether the flicker hides or destroys information is set by the write-time protector; an error-gated schedule can harvest hidden facts at no extra cost — to 85–96% of a day — but only while its self-tests are fresh. Where writes land matters for *expression* far more than for *storage*, and the recall advantage of late layers inverts the placement folklore of the batch regime. The structure replicates across two substrates even where the magnitudes do not. All of this is invisible to aggregate accuracy; the per-fact instrument is the contribution we most hope others reuse.

References

- Aljundi, R., et al. Online continual learning with maximally interfered retrieval. *NeurIPS* 2019.
- Allal, L. B., et al. SmolLM2: When smol goes big — data-centric training of a small language model. arXiv:2502.02737.
- Allen-Zhu, Z., & Li, Y. Physics of language models: Part 3.1, knowledge storage and extraction. *ICML* 2024.
- Back, S., Lee, D., Kang, N., Lee, T., Hong, S. K., Gwon, Y., & Ahn, S. Understanding LoRA as knowledge memory: An empirical analysis. *ICML* 2026. arXiv:2603.01097.
- Behrouz, A., Zhong, P., & Mirrokni, V. Titans: Learning to memorize at test time. arXiv:2501.00663.
- Benna, M. K., & Fusi, S. Computational principles of synaptic memory consolidation. *Nature Neuroscience*, 19(12).
- Fang, Y., Yu, W., Zhong, S., Ye, Q., Xiong, X., & Wei, L. Artificial hippocampus networks for efficient long-context modeling. arXiv:2510.07318.
- Farajtabar, M., et al. Orthogonal gradient descent for continual learning. *AISTATS* 2020.
- Hazard, H., Fountas, Z., Benfeghoul, M. A., Oomerjee, A., Wang, J., & Bou-Ammar, H. SuRe: Surprise-driven prioritised replay for continual LLM learning. arXiv:2511.22367.
- Hu, E., et al. LoRA: Low-rank adaptation of large language models. *ICLR* 2022.
- Kirkpatrick, J., et al. Overcoming catastrophic forgetting in neural networks. *PNAS*, 114(13).
- Kotha, S., Springer, J. M., & Raghunathan, A. Understanding catastrophic forgetting in language models via implicit inference. *ICLR* 2024. arXiv:2309.10105.

- Lu, Y., He, Y., Chen, J., & Zha, H. MSSR: Memory-aware adaptive replay for continual LLM fine-tuning. arXiv:2603.09892.
- McClelland, J. L., McNaughton, B. L., & O'Reilly, R. C. Why there are complementary learning systems in the hippocampus and neocortex. *Psych. Review*, 102(3).
- McCloskey, M., & Cohen, N. J. Catastrophic interference in connectionist networks. *Psychology of Learning and Motivation*, 24.
- Meng, K., Bau, D., Andonian, A., & Belinkov, Y. Locating and editing factual associations in GPT. *NeurIPS* 2022.
- Meng, K., et al. Mass-editing memory in a transformer. *ICLR* 2023.
- Robins, A. Catastrophic forgetting, rehearsal and pseudorehearsal. *Connection Science*, 7(2).
- Roediger, H. L., & Karpicke, J. D. Test-enhanced learning: Taking memory tests improves long-term retention. *Psychological Science*, 17(3).
- Schaul, T., Quan, J., Antonoglou, I., & Silver, D. Prioritized experience replay. *ICLR* 2016. arXiv:1511.05952.
- Toneva, M., et al. An empirical study of example forgetting during deep neural network learning. *ICLR* 2019.
- Xu, X., Yue, X., Liu, Y., Ye, Q., Zheng, H., Hu, P., Du, M., & Hu, H. Unlearning isn't deletion: Investigating reversibility of machine unlearning in LLMs. arXiv:2505.16831.
- Zhang, Z., Wang, F., Li, X., Wu, Z., Tang, X., Liu, H., He, Q., Yin, W., & Wang, S. Catastrophic failure of LLM unlearning via quantization. *ICLR* 2025. arXiv:2410.16454.
- Zhang, Z., Hu, R., Glentis, A., Li, D., Yau, C.-Y., Lin, H., & Hong, M. Is one layer enough? Training a single transformer layer can match full-parameter RL training. arXiv:2607.01232.

A Reproduction

Driver: `accum.py` (mechanism, `--layers`, `--replay-policy`, `--probe-every`, `--model`, recognition margins logged at every probe); `analyze.py` re-derives all tables from the shipped raw per-fact timelines (`results/accum_*.json`) with the pinned recovery and first-miss definitions; `reproduce.sh` is the full run matrix. Hyperparameters as in §3; seeds 1234/2025/777 (+7/42 for the $n=5$ compositions) throughout.